

CSA 0921 - PROGRAMMING IN JAVA

CO1 - AT3 - PSEUDO CODE WRITING

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ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

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CO1-AT3-PSEUDOCODE WRITING

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QUESTIONS

Q1

Student Management System (BL3 - Apply)
10 marks

✓

Q2

Library Book Management (BL3 - Apply)
10 marks

✓

Q3

Employee Salary Calculator (BL3 - Apply)
10 marks

✓

Q4

Simple Area Calculator using Polymorphism (BL3 - Apply)
10 marks

✓

Q5

Vehicle Registration System (BL3 - Apply)
10 marks

✓

Q6

Bank Account Hierarchy (BL3 - Analyze)
10 marks

✓

Q7

Hospital Patient Record System (BL3 - Analyze)
10 marks

✓

Q8

E-commerce Product Catalog (BL3 - Analyze)
10 marks

✓

Q9

University Staff Hierarchy (BL3 - Analyze)
10 marks

✓

Q10

Smart Home Device Controller (BL3 - Analyze)
10 marks

✓

10/10 answered

2. Student Management System (BL3 - Apply)
10 marks

Write pseudocode to design a Student class with attributes (name, rollNo, marks) and methods to:
• Accept student details
• Calculate average marks
• Display result (Pass/Fail)
Apply encapsulation and data hiding by restricting direct access to attributes.
OOP Concepts: Encapsulation | Data Hiding | Classes & Objects

Your Answer

START
CLASS Student
PRIVATE name
PRIVATE rollno
PRIVATE marks
PRIVATE average
METHOD acceptDetails()
INPUT name
INPUT rollno
INPUT marks
END METHOD
METHOD calculateAverage()
PRINT "Name: " + name
PRINT "Roll no.: " + rollno
PRINT "Marks: " + marks
PRINT "Average: " + average
IF average >= 50 THEN
PRINT "Result: Pass"
ELSE
PRINT "Result: Fail"
END IF
END METHOD
END CLASS
CREATE object s OF Student
CALL s.acceptDetails()
CALL s.calculateAverage()
CALL s.displayResult()
STOP

Save Answer

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QUESTIONS

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Q3

Employee Salary Calculator (BL3 - Apply)
10 marks

✓

Q4

Simple Area Calculator using Polymorphism (BL3 - Apply)
10 marks

✓

Q5

Vehicle Registration System (BL3 - Apply)
10 marks

✓

Q6

Bank Account Hierarchy (BL3 - Analyze)
10 marks

✓

Q7

Hospital Patient Record System (BL3 - Analyze)
10 marks

✓

Q8

E-commerce Product Catalog (BL3 - Analyze)
10 marks

✓

Q9

University Staff Hierarchy (BL3 - Analyze)
10 marks

✓

Q10

Smart Home Device Controller (BL3 - Analyze)
10 marks

✓

10/10 answered

2. Library Book Management (BL3 - Apply)
10 marks

Design a Book class with private attributes (Title, author, ISBN, isAvailable). Implement methods to:
• Issue a book (change availability status)
• Return a book
• Display book details using getter methods
Demonstrate object creation for at least 3 books and simulate issue/return operations.
OOP Concepts: Encapsulation | Getters & Setters | Objects

Your Answer

start
class book
private title
private author
private ISBN
private isAvailable
method getDetails() {
title ← t
author ← a
ISBN ← i
isAvailable ← true
}
end method
method issueBook() {
if isAvailable then
isAvailable ← false
print "Book issued"
else
print "Book is not available"
end if
}
end method
method returnBook() {
isAvailable ← true
print "Book returned"
}
end method
method displayDetails() {
print title
print author
print ISBN
print isAvailable
}
end method
end class
create objects book1, book2, book3 of book
call book1.setDetails("Java", "James", "101")
call book2.setDetails("Python", "Guido", "102")
call book3.setDetails("C++", "Bjarne", "103")
call book1.issueBook()
call book2.issueBook()
call book3.issueBook()
stop

Save Answer

C01-AT3-PSEUDOCODE WRITING

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QUESTIONS

- Q1 System Management System (B.L - Apply) 10 marks
- Q2 Library Book Management (B.L - Apply) 10 marks
- Q3 Employee Salary Calculation (B.L - Apply) 10 marks
- Q4 Shape Area Calculator using Polymorphism (B.L - Apply) 10 marks
- Q5 Vehicle Registration System (B.L - Apply) 10 marks
- Q6 Bank Account Hierarchy (B.L - Analyze) 10 marks
- Q7 Hospital Patient Record System (B.L - Analyze) 10 marks
- Q8 E-commerce Product Catalog (B.L - Analyze) 10 marks
- Q9 University Staff Hierarchy (B.L - Analyze) 10 marks
- Q10 Smart Home Device Controller (B.L - Analyze) 10 marks

10/10 answered

Employee Salary Calculation (B.L3 - Apply)

10 marks

Write pseudocode for an Employee class with encapsulated attributes (empId, name, basicPay, designation). Implement methods to:

- Calculate gross salary (basic + HRA + DA)
- Calculate net salary after deductions (PF, tax)
- Display pay slip

Apply data hiding to salary components are not directly accessible.

OOP Concepts: Encapsulation | Data Hiding | Methods

Your Answer

```
START
CLASS Employee
    PRIVATE empId, name, designation, basicPay
    PRIVATE HRA, DA, PF, tax, grossSalary, netSalary
    METHOD calculateGross()
    METHOD calculateNet()
    METHOD displayPaySlip()
END CLASS

CREATE object e
CALL e.calculateGross()
CALL e.calculateNet()
CALL e.displayPaySlip()
STOP
```

Save Answer

C01-AT3-PSEUDOCODE WRITING

← Back

QUESTIONS

- Q1 System Management System (B.L - Apply) 10 marks
- Q2 Library Book Management (B.L - Apply) 10 marks
- Q3 Employee Salary Calculation (B.L - Apply) 10 marks
- Q4 Shape Area Calculator using Polymorphism (B.L - Apply) 10 marks
- Q5 Vehicle Registration System (B.L - Apply) 10 marks
- Q6 Bank Account Hierarchy (B.L - Analyze) 10 marks
- Q7 Hospital Patient Record System (B.L - Analyze) 10 marks
- Q8 E-commerce Product Catalog (B.L - Analyze) 10 marks
- Q9 University Staff Hierarchy (B.L - Analyze) 10 marks
- Q10 Smart Home Device Controller (B.L - Analyze) 10 marks

10/10 answered

Shape Area Calculator using Polymorphism (B.L3 - Apply)

10 marks

Create a Shape base class with an overrideable calculateArea() method. Apply this to:

- Circle — area using radius
- Rectangle — area using length and breadth
- Triangle — area using base and height

Use method overriding to demonstrate runtime polymorphism. Call calculateArea() for each object.

OOP Concepts: Inheritance | Polymorphism | Method Overriding

Your Answer

```
START
CLASS Shape
    METHOD calculateArea()
    PRINT "Area"
END METHOD
END CLASS

CLASS Circle EXTENDS Shape
    radius
    OVERRIDE calculateArea()
    area = 3.14 * radius * radius
    PRINT area
END METHOD
END CLASS

CLASS Rectangle EXTENDS Shape
    length, breadth
    OVERRIDE calculateArea()
    area = length * breadth
    PRINT area
END METHOD
END CLASS

CLASS Triangle EXTENDS Shape
    base, height
    OVERRIDE calculateArea()
    area = 0.5 * base * height
    PRINT area
END METHOD
END CLASS

CREATE Circle, Rectangle, Triangle objects
CALL calculateArea() for each object
STOP
```

Save Answer

C01-AT3-PSEUDOCODE WRITING

10 marks

QUESTIONS

- Q1
Account Management System
(BL-1 - Apply)
10 marks
- Q2
Library Book Management
(BL-1 - Apply)
10 marks
- Q3
Employee Salary Calculation
(BL-1 - Apply)
10 marks
- Q4
Simple Area Calculator using
Polymorphism (BL-1 - Apply)
10 marks
- Q5
Vehicle Registration System
(BL-1 - Apply)
10 marks
- Q6
Bank Account Hierarchy (BL-1 - Analyze)
10 marks
- Q7
Hospital Patient Account System
(BL-1 - Analyze)
10 marks
- Q8
E-commerce Product Catalog
(BL-1 - Analyze)
10 marks
- Q9
University Staff Hierarchy (BL-1 - Analyze)
10 marks
- Q10
Smart Home Device Controller
(BL-1 - Analyze)
10 marks

10/10 answered

Vehicle Registration System (BL-1 - Apply)

10 marks

Design a Vehicle class with attributes (regNo, owner, fuelType). Extend it into:

- TwoWheeler — with engine capacity
- FourWheeler — with seating capacity and AC status

Apply inheritance to make common attributes and override a displayDetails() method in each subclass.

OOP Concepts: Inheritance | Method Overriding | Subclass

Your Answer

```
START
CLASS Vehicle
    regno, owner, fuelType
    METHOD displayDetails()
        PRINT regno, owner, fuelType
    END METHOD
END CLASS

CLASS TwoWheeler EXTENDS Vehicle
    engineCapacity
    OVERRIDE displayDetails()
        PRINT regno, owner, fuelType, engineCapacity
    END METHOD
END CLASS

CLASS FourWheeler EXTENDS Vehicle
    seatingCapacity, ACstatus
    OVERRIDE displayDetails()
        PRINT regno, owner, fuelType
        PRINT seatingCapacity, ACstatus
    END METHOD
END CLASS

CREATE TwoWheeler object
CREATE FourWheeler object
CALL displayDetails()
STOP
```

Save Answer

C01-AT3-PSEUDOCODE WRITING

10 marks

QUESTIONS

- Q1
Account Management System
(BL-1 - Apply)
10 marks
- Q2
Library Book Management
(BL-1 - Apply)
10 marks
- Q3
Employee Salary Calculation
(BL-1 - Apply)
10 marks
- Q4
Simple Area Calculator using
Polymorphism (BL-1 - Apply)
10 marks
- Q5
Vehicle Registration System
(BL-1 - Apply)
10 marks
- Q6
Bank Account Hierarchy (BL-1 - Analyze)
10 marks
- Q7
Hospital Patient Account System
(BL-1 - Analyze)
10 marks
- Q8
E-commerce Product Catalog
(BL-1 - Analyze)
10 marks
- Q9
University Staff Hierarchy (BL-1 - Analyze)
10 marks
- Q10
Smart Home Device Controller
(BL-1 - Analyze)
10 marks

10/10 answered

Bank Account Hierarchy (BL-1 - Analyze)

10 marks

Develop pseudocode for a BankAccount superclass and two subclasses — SavingsAccount (with interest calculation) and CurrentAccount (with overdraft facility). Analyze and implement:

- Inheritance of common attributes (acctNo, balance, holder)
- Method overriding for withdrawal behavior
- How overdraft protection differs from savings limit

Compare the behavior of withdraw() across both subclasses and justify the design choices.

OOP Concepts: Inheritance | Method Overriding | Polymorphism

Your Answer

```
START
CLASS BankAccount
    acctNo, holder, balance
    METHOD withdraw(amount)
        balance -= balance-amount
    END METHOD
END CLASS

CLASS SavingsAccount EXTENDS BankAccount
    OVERRIDE withdraw(amount)
        IF amount > balance THEN
            balance -= balance-amount
            PRINT "Insufficient Balance"
        END IF
    END METHOD
END CLASS

CLASS CurrentAccount EXTENDS BankAccount
    overdraftLimit
    OVERRIDE withdraw(amount)
        IF amount > balance+overdraftLimit THEN
            balance -= balance-amount
            PRINT "Overdraft Limit Exceeded"
        END IF
    END METHOD
END CLASS

CREATE SavingsAccount object
CREATE CurrentAccount object
CALL withdraw()
STOP
```

Save Answer

C01-AT3-PSEUDOCODE WRITING

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QUESTIONS

- Q1 Student Management System (BL4 - Analyze) 10 marks
- Q2 Library Book Management (BL4 - Analyze) 10 marks
- Q3 Employee Salary Calculation (BL4 - Analyze) 10 marks
- Q4 Shop Area Calculator using Inheritance (BL4 - Analyze) 10 marks
- Q5 Vehicle Registration System (BL4 - Analyze) 10 marks
- Q6 Bank Account Hierarchy (BL4 - Analyze) 10 marks
- Q7 Hospital Patient Record System (BL4 - Analyze) 10 marks
- Q8 E-Commerce Product Catalog (BL4 - Analyze) 10 marks
- Q9 University Staff Hierarchy (BL4 - Analyze) 10 marks
- Q10 Retail Store Device Controller (BL4 - Analyze) 10 marks

10/10 answered

Hospital Patient Record System (BL4 - Analyze)

10 marks

Design a Patient base class and analyze how it extends into:

- InPatient — with ward number, admission date, and daily charges
- OutPatient — with appointment date and consultation fee

Analyze encapsulation requirements for sensitive data (diagnosis, prescriptions). Override a generateBill() method for each type and compare the billing logic differences. OOP Concepts: Encapsulation | Inheritance | Polymorphism

Your Answer

```
START
CLASS Patient
    PRIVATE patientId, name, diagnosis, prescription
    METHOD generateBill()
        PRINT "Patient Bill"
    END METHOD
END CLASS

CLASS InPatient EXTENDS Patient
    wardNo, admissionDate, dailyCharges
    OVERRIDE generateBill()
        bill = dailyCharges * days
        PRINT bill
    END METHOD

END CLASS

CLASS OutPatient EXTENDS Patient
    appointmentDate, consultationFee
    OVERRIDE generateBill()
        PRINT consultationFee
    END METHOD

END CLASS

CREATE InPatient object
CREATE OutPatient object
CALL generateBill()
STOP
```

Save Answer

C01-AT3-PSEUDOCODE WRITING

Back

QUESTIONS

- Q1 Student Management System (BL4 - Analyze) 10 marks
- Q2 Library Book Management (BL4 - Analyze) 10 marks
- Q3 Employee Salary Calculation (BL4 - Analyze) 10 marks
- Q4 Shop Area Calculator using Inheritance (BL4 - Analyze) 10 marks
- Q5 Vehicle Registration System (BL4 - Analyze) 10 marks
- Q6 Bank Account Hierarchy (BL4 - Analyze) 10 marks
- Q7 Hospital Patient Record System (BL4 - Analyze) 10 marks
- Q8 E-Commerce Product Catalog (BL4 - Analyze) 10 marks
- Q9 University Staff Hierarchy (BL4 - Analyze) 10 marks
- Q10 Retail Store Device Controller (BL4 - Analyze) 10 marks

10/10 answered

E-Commerce Product Catalog (BL4 - Analyze)

10 marks

Analyze and design a Product class hierarchy for an e-commerce system:

- Electronics — with warranty, brand, voltage
- Clothing — with size, fabric, gender
- Grocery — with expiryDate and weight

Override an applyDiscount() method for each category with different discount rules. Analyze why polymorphism makes the cart checkout process flexible and extensible. OOP Concepts: Polymorphism | Inheritance | Method Overriding

Your Answer

```
START
CLASS Product
    productId, name, price
    METHOD applyDiscount()
        PRINT price
    END METHOD
END CLASS

CLASS Electronics EXTENDS Product
    OVERRIDE applyDiscount()
        price = price * (price < 100)
        PRINT price
    END METHOD

END CLASS

CLASS Clothing EXTENDS Product
    OVERRIDE applyDiscount()
        price = price * (price < 200)
        PRINT price
    END METHOD

END CLASS

CLASS Grocery EXTENDS Product
    OVERRIDE applyDiscount()
        price = price * (price < 50)
        PRINT price
    END METHOD

END CLASS

CREATE Electronics, Clothing, Grocery objects
CALL applyDiscount()
STOP
```

Save Answer

C01-AT3-PSEUDOCODE WRITING

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QUESTIONS

- Q1
Student Management System
(BLA - Apply)
10 marks
- Q2
2-Library Book Management
(BLA - Apply)
10 marks
- Q3
Employee Salary Calculation
(BLA - Apply)
10 marks
- Q4
Hotel Area Calculation using
Polymorphism (BLA - Apply)
10 marks
- Q5
Vehicle Registration System
(BLA - Apply)
10 marks
- Q6
Bank Account Hierarchy (BLA -
Analyze)
10 marks
- Q7
Hospital Patient Record System
(BLA - Analyze)
10 marks
- Q8
E-Commerce Product Catalog
(BLA - Analyze)
10 marks
- Q9
University Staff Hierarchy (BLA -
Analyze)
10 marks
- Q10
Smart Home Device Controller
(BLA - Analyze)
10 marks

10/10 answered

University Staff Hierarchy (BLA - Analyze)

10 marks

Analyze the design of a Staff base class extended by TeachingStaff and NonTeachingStaff. Further extend TeachingStaff into Professor and Lecturer (multilevel inheritance). For each level:

- Identify which attributes should be private vs protected
- Overload calculateAllowance() at each level
- Analyze how protected access enables inheritance without full exposure. Justify the use of multilevel inheritance and explain how access modifiers enforce data hiding.

OOP Concepts: Multilevel Inheritance | Access Modifiers | Data Hiding

Your Answer

```
START
CLASS Staff
    PRIVATE staffId, name
    PROTECTED salary
    METHOD calculateAllowance()
        PRINT "Allowance"
    END METHOD
END CLASS
CLASS TeachingStaff EXTENDS Staff
    OVERRIDE calculateAllowance()
        PRINT salary*0.20
    END METHOD
END CLASS
CLASS Professor EXTENDS TeachingStaff
    OVERRIDE calculateAllowance()
        PRINT salary*0.30
    END METHOD
END CLASS
CLASS Lecturer EXTENDS TeachingStaff
    OVERRIDE calculateAllowance()
        PRINT salary*0.25
    END METHOD
END CLASS
CLASS NonTeachingStaff EXTENDS Staff
    OVERRIDE calculateAllowance()
        PRINT salary*0.15
    END METHOD
END CLASS
CREATE Professor, Lecturer, NonTeachingStaff objects
CALL calculateAllowance()
STOP
```

Save Answer

C01-AT3-PSEUDOCODE WRITING

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QUESTIONS

- Q1
Student Management System
(BLA - Apply)
10 marks
- Q2
2-Library Book Management
(BLA - Apply)
10 marks
- Q3
Employee Salary Calculation
(BLA - Apply)
10 marks
- Q4
Hotel Area Calculation using
Polymorphism (BLA - Apply)
10 marks
- Q5
Vehicle Registration System
(BLA - Apply)
10 marks
- Q6
Bank Account Hierarchy (BLA -
Analyze)
10 marks
- Q7
Hospital Patient Record System
(BLA - Analyze)
10 marks
- Q8
E-Commerce Product Catalog
(BLA - Analyze)
10 marks
- Q9
University Staff Hierarchy (BLA -
Analyze)
10 marks
- Q10
Smart Home Device Controller
(BLA - Analyze)
10 marks

10/10 answered

Smart Home Device Controller (BLA - Analyze)

10 marks

Analyze and design a SmartDevice superclass with subclasses: SmartLight, SmartThermostat, and SmartLock. Override an executeCommand(cmd) method in each, where the same command (e.g., 'on', 'off', 'status') produces device-specific behavior. Analyze:

- How polymorphism allows a single controller loop to manage all devices
- Security implications of data hiding in SmartLock (PIN must never be exposed) Design the class hierarchy and explain the OOP principles applied at each level.

OOP Concepts: Polymorphism | Encapsulation | Inheritance

Your Answer

```
START
CLASS SmartDevice
    METHOD executeCommand(cmd)
        PRINT cmd
    END METHOD
END CLASS
CLASS SmartLight EXTENDS SmartDevice
    OVERRIDE executeCommand(cmd)
        PRINT "Light" + cmd
    END METHOD
END CLASS
CLASS SmartThermostat EXTENDS SmartDevice
    OVERRIDE executeCommand(cmd)
        PRINT "Thermostat" + cmd
    END METHOD
END CLASS
CLASS SmartLock EXTENDS SmartDevice
    PRIVATE PIN
    OVERRIDE executeCommand(cmd)
        PRINT "Lock" + cmd
    END METHOD
END CLASS
CREATE SmartLight object
CREATE SmartThermostat object
CREATE SmartLock object
CALL executeCommand("ON")
CALL executeCommand("OFF")
CALL executeCommand("STATUS")
STOP
```

Save Answer